# Exploiting Malicious RIS for Secret Key Acquisition in Physical-Layer Key Generation

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***Abstract*—Due to the capability of controlling wireless environments, reconfigurable intelligent surface (RIS) can be exploited to assist radio channel-based physical-layer key generation (PKG). However, this may also pose great security threats to PKG when it is controlled by an attacker. To this end, this letter studies a RIS leakage (RISL) attack that assists the attacker to manipulate the generated keys between the legitimate ends in PKG systems. Specifically, in block fading environments, by alternatingly adapting the RIS reflection coefficients to all 0s and 1s in each block, the received signal strength (RSS) at the legitimate ends is likely to be boosted and attenuated regularly across time, causing predictable 0s and 1s in the quantized bits. To resist this attack, we propose a countermeasure based on dynamic private pilots (CDPP) and prove that the RISL success probability can be significantly reduced by CDPP. Simulation results verify the effectiveness of the proposed attack and countermeasure, highlighting the importance of introducing artificial randomness for secret key security in block fading channels.**

***Index Terms*—Physical layer security, secret key generation, reconfigurable intelligent surface (RIS), active attack.**

## I. INTRODUCTION

PHYSICAL-LAYER key generation (PKG) is a promising technique to establish symmetric keys between legitimate ends in wireless communication networks [1]. By leveraging the inherent randomness and reciprocity of wireless channels, two legitimate ends, i.e., Alice and Bob, respectively, can extract a pair of secret keys from their shared common channels. However, since the performance of PKG is determined by the properties of wireless channels, the desired key generation performance cannot be always guaranteed in some hash environments due to wave-blocks [2].

Fortunately, in the past five years, reconfigurable intelligent surface (RIS) has emerged as a disruptive wireless communication technology that enables customization of wireless propagation environments [3]. Indeed, RIS is a planar surface that consists of a large number of passive reflecting elements, which can be altered to collaboratively establish favourable wireless channels. Therefore, RIS can be applied in PKG to possibly enhance the key generation performance. As such, some of the existing works have investigated RIS-assisted PKG schemes. In particular, when RIS is controlled by some legitimate parties, it can harness the temporal fluctuation of the channel realizations or serve as a new degrees-of-freedom (DoF) for optimizing the key generation rate (KGR). Specifically, in static environments, [4], [5], [6], [7], [8] randomly altered the phase of each RIS element to increase the channel randomness so as to improve the entropy of secret keys. Besides, [9], [10] optimized the RIS configuration to maximize the KGR in different dynamic environments.

Despite these attractive performance improvements, RIS is also a double-edged sword that may also bring potential attacks against PKG. In fact, when a RIS is somehow controlled by a malicious attacker, Eve, it may jeopardize the PKG process or even help Eve obtain the secret key information. Specifically, an attacker in [11] adapted the RIS phases in the uplink and downlink to destroy the channel reciprocity between the legitimate parties. To resist this attack, a path separation-based countermeasure was proposed to detect this attack and improve the KGR. Besides, since the end-to-end channel between the legitimate parties is the superimposition of the direct link and the RIS-induced link, Eve in [12] adopted a semi-passive RIS exploiting the compressive sensing technique to estimate the RIS-induced link such that part of the secret key can be obtained. In this letter, instead of estimating the RIS-induced link, we focus on a scenario that an attacker launches a RIS leakage (RISL) attack to influence the generated bits at the legitimate ends by adapting the RIS reflection matrix in a planned manner. The initial idea of RISL was partly presented in our previous magazine [2], but the crucial implementation of the attack and countermeasure remains unknown. More importantly, the corresponding performance analysis is still lacked. Specifically, the main contributions of this letter are as follows.

- We study a new RIS-based active attack, RISL, in PKG systems. We show that in block fading environments,

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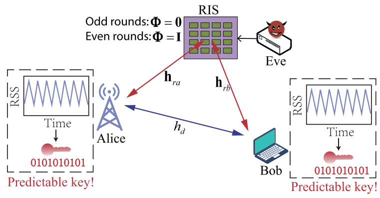


Fig. 1. System model of the RISL attack.

by alternatingly setting the reflection coefficients to all 0s and 1s in each block, the received signal strength (RSS) can be boosted and attenuated regularly following a certain pattern across time, resulting in predictable secret keys at the legitimate ends.

- We then derive the probability of Eve in successfully obtaining the secret key, which is decided by the power ratio of the end-to-end channel to the direct link. To resist this attack, we also propose a countermeasure based on dynamic private pilots (CDPP). We prove that the success probability is greatly reduced and is upper bounded.
- Simulation results verify that when the RIS-induced link dominates the end-to-end superimposed channel, the success probability of the RISL attack approaches 1. In contrast, the success probability of the attack in the CDPP can be reduced substantially and coincide with the derived theoretical analysis.

## II. System Model and Problem Formulation

As shown in Fig. 1, we consider a RSS-based PKG model, where two single-antenna legitimate ends, i.e., Alice and Bob, aim to generate a symmetric key by exploiting their reciprocal RSS[1] in time-division duplexing (TDD) mode. Meanwhile, an active attacker, Eve, is equipped with a reflective RIS consisting of $N$ passive reflection elements [3], [4], [5]. By adapting the RIS reflection coefficients, Eve aims to alter the wireless propagation channel to influence the generated secret keys between Alice and Bob.

### A. RIS-Involved Channel Model

In this letter, we consider a block fading environment [1], where all channel coefficients remain constant in one block. We assume each block is sufficiently long and Alice and Bob perform key generation within it. The direct link between Alice and Bob is denoted as $h_d \in \mathbb{C}$, where $\mathbb{C}^{A \times B}$ denotes the space of complex matrices of size $A \times B$. When the RIS is introduced to the PKG system, it establishes some indirect RIS-induced links. Specifically, the channel from the RIS to Alice and that to Bob are denoted as $\mathbf{h}_{ra} \in \mathbb{C}^{N\times 1}$ and $\mathbf{h}_{rb} \in \mathbb{C}^{N\times 1}$, respectively. Meanwhile, the RIS reflection matrix alters its coefficients over time. Specifically, at time $t(i) = iT_s, i \in \{1, \ldots, L\}$, where $T_s$ is the period of the repeated bidirectional channel probings of the legitimate ends and $L$ is the number of channel probing rounds, the reflection matrix of the RIS is denoted by the diagonal matrix $\boldsymbol{\Phi}(i) \in \mathbb{C}^{N\times N}$, where its diagonal elements are the reflection coefficients. In this case, the equivalent channel between Alice and Bob at time $t(i)$ is[2]

$$h_{ba}(i) = \mathbf{h}_{rb}^T \boldsymbol{\Phi}(i)\, \mathbf{h}_{ra} + h_d, \forall i, \tag{1}$$

where $(\cdot)^T$ denotes the transpose.

### B. PKG Process

The general PKG process consists of four phases [1]. First, during the channel probing phase, Alice and Bob alternatingly probe the channel to extract their similar channel characteristics. Specifically, in the $i$-th round of the probing, Alice and Bob transmit pilot $s \in \mathbb{C}$ with $|s|^2 = 1$, where $|\cdot|$ denotes the modulus of a complex number. Then, the received signal at Bob and Alice is denoted as $y_u(i) = h_{ba}(i)s + n_u(i), u \in \{a, b\}$, respectively, where $n_u(i)$ is the complex Gaussian noise with zero mean and variance $\sigma_n^2$. Since the public pilot $s$ is known by Alice, Bob, and Eve, we assume $s = 1$ without loss of generality [13]. Then, the RSS at the legitimate parties is calculated as $\mathrm{RSS}_u(i) = |y_u(i)|^2$.

After $L$ rounds of channel probing, the legitimate parties obtain a RSS sequence, $\{\mathrm{RSS}_u(1), \ldots, \mathrm{RSS}_u(L)\}$. Next, the mean value-based quantization technique is commonly adopted to convert it into a binary sequence [1]. Specifically, the legitimate parties calculate the mean value as $\overline{\mathrm{RSS}}_u = \mathbb{E}\{\mathrm{RSS}_u(i)\}, \forall u$, where $\mathbb{E}\{\cdot\}$ represents statistical expectation. Then, the quantized bits are decided by

$$K_u(i) = \begin{cases} 1, & \mathrm{RSS}_u(i) \geq \overline{\mathrm{RSS}}_u \\ 0, & \mathrm{RSS}_u(i) < \overline{\mathrm{RSS}}_u, \end{cases} \forall i. \tag{2}$$

Information reconciliation and privacy amplification are finally adopted to convert the quantized bits into the secret key [1].

### C. Problem Formulation

Note that since the information reconciliation and privacy amplification algorithms are public, the security of the key is guaranteed by the quantized bits $K_u$. Therefore, if Eve could obtain $K_u$, it implies that the secret key is also acquired by Eve [2], [7], [12]. Also, from the channel model in (1), the variation of the RSS is impacted by the reflection matrix $\boldsymbol{\Phi}(i)$. Thus, Eve can adapt $\boldsymbol{\Phi}(i)$ over time to generate controllable channel fluctuation favouring its attack. Inspired by this, we next propose a RISL attack launched by Eve that manipulates $\boldsymbol{\Phi}(i)$ regularly to produce predictable secret key bits.

## III. RISL Attack

In this section, we first describe the process of the RISL attack and then evaluate its performance.

### A. Attack Description

The key step of RISL involves setting the RIS reflection coefficients to all 0s and 1s in odd and even number rounds, respectively. This deliberated manipulation causes periodic signal enhancement and attenuation of the RSS. Specifically,

[1]RSS is one of the most popular characteristics for key generation. It has been exploited in almost all practical wireless standards, such as IEEE 802.11, IEEE 802.15.4, bluetooth, and LoRa, etc. [1].

[2]The adopted RIS-involved channel model (1) can be applied to both the single RIS and multiple RISs cases.

during the odd rounds, the reflection matrices are $\mathbf{0}$ and the received signals at the legitimate parties are

$$y_u(2\ell-1) = h_d + n(2\ell-1), \ell \in \{1, \ldots, L/2\}. \tag{3}$$

Then, during the even rounds, the reflection matrix is an unitary matrix, $\mathbf{I}$, and the received signals at the legitimate parties are

$$y_u(2\ell) = h_r + h_d + n(2\ell), \tag{4}$$

where $h_r \triangleq \mathbf{h}_{rb}^T \mathbf{h}_{ra}$ denotes the RIS-induced link.

From (3) and (4), the received signals in the even rounds contain both the RIS-induced link $h_r$ and the direct link $h_d$, while the received signals in the odd rounds only contain $h_d$. Hence, it is possible that the RSSs of the even rounds are stronger than that of the odd rounds, especially when the magnitude of $h_r$ is much larger than $h_d$. Thus, Eve predicts the generated bits sequence as $K_e = \{0, 1, \ldots, 0, 1\}$ and the RISL attack is considered successful if the predicted sequence at Eve matches exactly the generated bits at the legitimate parties, i.e., $K_u = K_e$.

### B. Attack Performance Analysis

To evaluate the attack performance, we derive the success probability of the RISL attack, which is summarized in the following theorem.

*Theorem 1:* The success probability of the RISL attack is

$$P_{\text{success}} = \mathbb{P}(K_u = K_e) \tag{5}$$

$$= \left[ e^{-|h_d|^2} \sum_{j=0}^{\infty} \frac{\left(|h_d|^2\right)^j}{j!} F_{2+2j}(x_0) \right]^{\frac{L}{2}} \times \left[ 1 - e^{-|h_d+h_r|^2} \sum_{j=0}^{\infty} \frac{\left(|h_d+h_r|^2\right)^j}{j!} F_{2+2j}(x_0) \right]^{\frac{L}{2}}, \tag{6}$$

where $x_0 = |h_d|^2 + |h_d + h_r|^2 + 2\sigma_n^2$, $F_q(x)$ is a cumulative distribution function of the central chi-squared distribution with $q$ DoF.

*Proof:* Please see Appendix A. ■

It can be observed that $P_{\text{success}}$ decreases with $L$. Indeed, as the sequence becomes longer, the randomness introduced by noise is likely to cause occasional irregular bits, which reduces the success probability. To provide further insights, we next analyze the success probability at the high SNR regime and the impact of noise will also be discussed via numerical results in Section V. Specifically, when $\sigma_n^2 \to 0$, the success probability of RISL is given by

$$P_{\text{success}}^{\text{high-SNR}} = \left( \mathbb{P}\left\{ |h_d + h_r|^2 / |h_d|^2 > 1 \right\} \right)^L. \tag{7}$$

*Remark 1:* From (7), the success probability is decided by the power ratio of the end-to-end composited channel to the direct link. Specifically, when $|h_d+h_r|^2 > |h_d|^2$, which means the RSSs in the even rounds are stronger than that of the odd rounds with the introduction of RIS. As a result, the generated key bits are close to Eve's prediction at high SNR such that the success probability approaches 1. By contrast, when $|h_d + h_r|^2 < |h_d|^2$, i.e., the introduction of RIS reduces the RSS in the even rounds, the generated bits sequence is likely to be $\{1, 0, \ldots, 1, 0\}$, which is completely different from Eve's prediction. In this case, the success probability approaches 0 with the noise variance decreases.

*Remark 2:* Assume that Eve has no information about the direct link $h_d$. To increase the ratio $|h_d + h_r|^2/|h_d|^2$, Eve would increase $h_r$ or decrease $h_d$ as much as possible. Specifically, Eve could take the following actions.

1) Increase the number of RIS elements or employ some power amplifiers. When there are more elements at the RIS, the RIS-induced link, $h_r$, contains more sub-channels corresponding to RIS elements, thereby increasing the magnitude of $h_r$. Also, the use of an active amplifier can enlarge the gain of $h_r$.
2) Deploy RIS close to either Alice or Bob. Since the large-scale fading of the RIS-induced link is the product of the gain of the Alice-RIS link and that of the RIS-Bob link, the gain of $h_r$ reaches its maximum when the RIS is located close either to Alice or Bob. Also, for the scenario involving mobile users, deploying multiple RISs could enhance the attack performance since the nearest RIS can provide stronger channel gains when the user moves to different locations.
3) Block the direct link between the legitimate ends. As $h_d \to 0$, we have $|h_d + h_r|^2/|h_d|^2 \to \infty$ and the success probability of the attack is close to 1. Indeed, in wave-blockage environments, the end-to-end channel is completely controlled by the RIS-induced link and Eve can easily manipulate the generated key bits.

Note that compared to the existing predictable channel attack in [14], the considered RISL only adapts the reflection matrices regularly over time, making it more undetectable and easier to implement. Also, since RISL does not require rapid variations of the reflection coefficients, as the interval of changes is the period of probing rounds $T_s$, the slewing rate detection method in [11] cannot be applied to detect RISL. Next, we propose a CDPP method to mitigate the impact of RISL via channel randomization.

## IV. CDPP Method: Countermeasure to Randomize the Channel

To resist the RISL attack, in this section we introduce a CDPP method and analyze the corresponding performance.

### A. Countermeasure Description

The basic idea of the CDPP method is to adopt artificial randomness to establish time-varying cascaded channels such that the variations of the channel are irregular. Specifically, Alice first generates an artificial randomness coefficient $w(i) \in \mathbb{C}$, which follows complex Gaussian distribution with zero mean and unit variance. Then, Alice transmits the private pilot $w(i)s$ and the received signal at Bob is expressed as

$$y_b(i) = h_{ba}(i)w(i)s + n_b(i). \tag{8}$$

Next, Bob sends $s$ and the received signal at Alice is

$$\bar{y}_a(i) = h_{ba}(i)s + \bar{n}_a(i). \tag{9}$$

Since the received signals, $y_b(i)$ and $\bar{y}_a(i)$, are quite different, Alice multiplies $\bar{y}_a(i)$ with $w(i)$ to obtain

$$y_a(i) = w(i)\bar{y}_a(i) = w(i)h_{ba}(i)s + n_a(i), \quad (10)$$

where $n_a(i) = \bar{n}_a(i)w(i)$.

From (8) and (10), the reciprocal channel gain in the CDPP, $w(i)h_{ba}(i)$, not only depends on the RIS reflection matrix $\mathbf{\Phi}(i)$ in $h_{ba}(i)$, but is related to the artificial randomness $w(i)$.[3] Hence, it can effectively counteract the attack by the RIS-assisted attacker for reducing the probability of regular RSS occurrences.[4]

### B. CDPP Performance Analysis

Since the received signal $y_u(i)$, contains two random variables, i.e., $w(i)$ and $n_u(i)$, it is difficult to derive a closed-form expression of $P_{\text{success}}$. To shed light on this, we next provide the success probability of the attack in the CDPP method at high SNR regime.

*Theorem 2:* At high SNR, the success probability of the RISL attack in the CDPP is expressed as

$$P_{\text{success}}^{\text{high-SNR}} = \left[F_2\left(|h_d + h_r|^2/|h_d|^2 + 1\right)\right]^{\frac{L}{2}} \times \left[1 - F_2\left(|h_d|^2/|h_d + h_r|^2 + 1\right)\right]^{\frac{L}{2}}. \quad (11)$$

*Proof:* Please see Appendix B. ∎

*Remark 3:* In (11), the success probability increases with $|h_d + h_r|^2/|h_d|^2$, since $F_2(|h_d + h_r|^2/|h_d|^2 + 1)$ increases while $F_2(|h_d + h_r|^2/|h_d|^2 + 1)$ decreases. This means the success probability still increases with the power ratio of the end-to-end channel to the direct link. However, when $|h_d + h_r|^2/|h_d|^2 \to \infty$, the upper bound of the success probability is

$$P_{\text{success}}^{\text{UB}} = \left(\mathbb{P}\left\{|w(2\ell)|^2 > 1/2\right\}\right)^{\frac{L}{2}} \approx 0.6065^{\frac{L}{2}}. \quad (12)$$

By comparing (7) and (12), we can observe that under an RISL attack without countermeasures, the success probability of the attack reaches 1 as long as $|h_d + h_r|^2 > |h_d|^2$. By contrast, with the application of the proposed CDPP method, the success probability is kept below $0.6065^{\frac{L}{2}}$ regardless the given values of $|h_d + h_r|^2$ and $|h_d|^2$. This verifies the effectiveness of the proposed countermeasure.

## V. Simulation Results

In this section, we verify the effects of the RISL attack and the CDPP method with Monte Carlo simulations.

First, Fig. 2 shows the success probability of the RISL attack versus $|h_d + h_r|^2/|h_d|^2$. It can be seen that the success probability increases monotonically and eventually reaches 1 for all the considered SNR values, since the RSS in the even rounds of probing is likely to be greater than that in the odd rounds with a larger $|h_d + h_r|^2/|h_d|^2$. Thus, the theoretical secret key rate (SKR) is 0, as all channel coefficients remain constant for a long time, allowing Eve to entirely predict the generated key. Furthermore, it can be seen that when $|h_d + h_r|^2/|h_d|^2 < 1$, the success probability decreases with the increase of SNR and approaches 0 rapidly when SNR $= 25$ dB, while it increases with the SNR when $|h_d + h_r|^2/|h_d|^2 > 1$. This is because as the noise variance decreases, the randomness caused by the noise is reduced. Then, the generated bits are more likely to align with Eve's prediction when $|h_d + h_r|^2/|h_d|^2 > 1$, while they tend to be entirely different from Eve's prediction for $|h_d + h_r|^2/|h_d|^2 < 1$. This observation is consistent with (7).

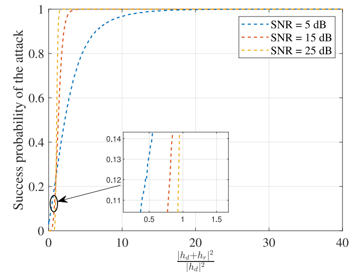


Fig. 2. The success probability of the RISL attack, $L = 2$.

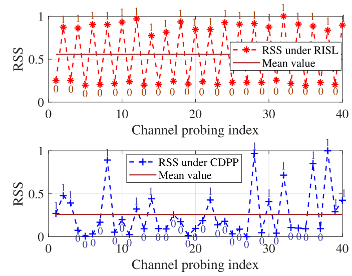


Fig. 3. Waveform of Alice under the RISL attack and the CDPP method, SNR $= 25$ dB, $|h_d + h_r|^2/|h_d|^2 = 4$, $L = 40$.

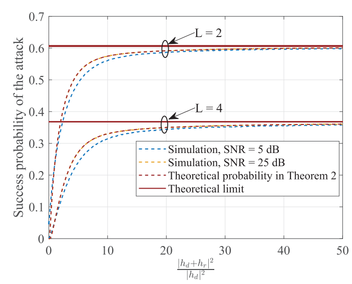


Fig. 4. The success probability of the RISL attack in the CDPP.

Then, Fig. 3 shows the RSSs of 40 channel probings instants in RISL and CDPP. It can be seen that the RSS varies regularly in RISL and thus, the quantized bits are alternating between 0s and 1s. In contrast, the RSS in the CDPP shows irregular variations due to the introduction of artificial randomness $w(i)$. As predicted, the quantized bits exhibit strong randomness and are unpredictable for Eve.

Finally, Fig. 4 shows the success probability of RISL in the CDPP. First, it can be seen that the success probability

[3]When Eve adopts a semi-passive/active RIS to estimate the RIS-induced link [12], the generated key is still challenging for Eve to predict, since the direct link between Alice and Bob is dynamic and cannot be acquired by Eve.

[4]Note that the proposed method is different from the one adopted in covert communication [15], since in this letter, the artificial randomness is exploited as secret key information, which, however, is utilized to hide information in covert communication.

still increases with $|h_d + h_r|^2/|h_d|^2$, since the RIS-induced link dominates the end-to-end channel. However, compared with the attack success probability close to 1, as shown in Fig. 2, for $L = 2$ and $L = 4$, the success probabilities are less than 0.6065 and $0.6065^2$ and approach them asymptotically for large $|h_d + h_r|^2/|h_d|^2$, respectively, which is in agreement with the derived upper bound in (12). Also, by employing the ITE toolbox [16], the minimum SKR of the CDPP for SNR $=$ 25 dB is 0.147 bits/channel use, which is greater than that of the RISL attack. Additionally, as can be observed, the success probability for SNR $=$ 25 dB coincides with the value of $P_{\text{success}}^{\text{high-SNR}}$ in Theorem 2. These results validate the theoretical analysis and demonstrate that the proposed CDPP can significantly reduce the success probability of RISL.

## VI. Conclusion

In this letter, we studied a RISL attack in PKG systems. In this attack, the attacker-controlled RIS varies its reflection coefficients deliberately to cause predictable alteration of the RSS across time. Hence, the generated bit sequence from these RSSs follows a certain designed pattern such that it is predictable. To mitigate the impact of RISL, we proposed a CDPP method and proved its effectiveness by deriving the upper bound of the success probability in the CDPP. Numerical results verified that under the RISL attack, Eve could predict the secret key when the strength of the RIS-induced link dominates the end-to-end composited communication channel. Meanwhile, the derived upper bound was validated through simulations, demonstrating the effectiveness of the proposed countermeasure.

## Appendix A
## Proof of Theorem 1

First, after $L$ rounds of channel probing, the mean of the RSSs at Alice and Bob are respectively calculated as $\overline{\text{RSS}}_u = \frac{1}{2}(|h_d|^2 + |h_d + h_r|^2) + \sigma_n^2, u \in \{a, b\}$. Thus, the success probability of RISL attack is expressed as

$$P_{\text{success}} = \left(\mathbb{P}\left\{\text{RSS}_u(2\ell-1) < \overline{\text{RSS}}_u, \text{RSS}_u(2\ell) > \overline{\text{RSS}}_u\right\}\right)^{\frac{L}{2}} \tag{13}$$

$$\overset{(a)}{=} \left(\mathbb{P}\left\{\text{RSS}_u(2\ell-1) < \overline{\text{RSS}}_u\right\}\right)^{\frac{L}{2}} \times\left(\mathbb{P}\left\{\text{RSS}_u(2\ell) > \overline{\text{RSS}}_u\right\}\right)^{\frac{L}{2}}, \tag{14}$$

where $(a)$ holds since the noise in different RSSs is independent. The first term in (14) is calculated as

$$\mathbb{P}\left\{\overline{\text{RSS}}_u - \text{RSS}_u(2\ell-1) > 0\right\} \tag{15}$$

$$= \mathbb{P}\Big\{2(\Re\{h_d + n_u(2\ell-1)\})^2 + 2(\Im\{h_d + n_u(2\ell-1)\})^2 < |h_d|^2 + |h_d + h_r|^2 + 2\sigma_n^2\Big\}. \tag{16}$$

Since $2(\Re\{h_d + n_u(2\ell-1)\})^2 + 2(\Im\{h_d + n_u(2\ell-1)\})^2$ follows a noncentral chi-squared distribution with 2 DoF and the noncentrality parameter being $\lambda = 2|h_d|^2$ [17], where $\Re\{\cdot\}$ and $\Im\{\cdot\}$ stand for the real and imaginary parts of a complex number, respectively, the probability (16) is calculated as the first term in (6) while its second term can be derived similarly. This completes the proof.

## Appendix B
## Proof of Theorem 2

As $\sigma_n^2 \to 0$, the success probability is expressed as

$$P_{\text{success}}^{\text{high}-\text{SNR}} = \left(\mathbb{P}\Big\{\Big(1 - 2|w(2\ell-1)|^2\Big)|h_d|^2 + |h_d + h_r|^2 > 0\Big\}\right)^{\frac{L}{2}} \times\left(\mathbb{P}\Big\{\Big(2|w(2\ell)|^2 - 1\Big)|h_d + h_r|^2 - |h_d|^2 > 0\Big\}\right)^{\frac{L}{2}} \tag{17}$$

$$= \left(\mathbb{P}\Big\{2|w(2\ell-1)|^2 < |h_d + h_r|^2/|h_d|^2 + 1\Big\}\right)^{\frac{L}{2}} \times\left(\mathbb{P}\Big\{2|w(2\ell)|^2 > |h_d|^2/\Big(|h_d + h_r|^2\Big) + 1\Big\}\right)^{\frac{L}{2}}. \tag{18}$$

Since $2|w(\ell)|^2$ follows a central chi-squared distribution with 2 DoF, we can obtain (11). This completes the proof.

## References


[1] J. Zhang, G. Li, A. Marshall, A. Hu, and L. Hanzo, "A new frontier for IoT security emerging from three decades of key generation relying on wireless channels," *IEEE Access*, vol. 8, pp. 138406–138446, 2020.

[2] G. Li et al., "Reconfigurable intelligent surface for physical layer key generation: Constructive or destructive?" *IEEE Wireless Commun. Mag.*, vol. 29, no. 4, pp. 146–153, Aug. 2022.

[3] S. Hu, Z. Wei, Y. Cai, C. Liu, D. W. K. Ng, and J. Yuan, "Robust and secure sum-rate maximization for multiuser MISO downlink systems with self-sustainable IRS," *IEEE Trans. Commun.*, vol. 69, no. 10, pp. 7032–7049, Oct. 2021.

[4] Z. Ji et al., "Random shifting intelligent reflecting surface for OTP encrypted data transmission," *IEEE Wireless Commun. Lett.*, vol. 10, no. 6, pp. 1192–1196, Jun. 2021.

[5] X. Hu, L. Jin, K. Huang, X. Sun, Y. Zhou, and J. Qu, "Intelligent reflecting surface-assisted secret key generation with discrete phase shifts in static environment," *IEEE Wireless Commun. Lett.*, vol. 10, no. 9, pp. 1867–1870, Sep. 2021.

[6] T. Lu, L. Chen, J. Zhang, K. Cao, and A. Hu, "Reconfigurable intelligent surface assisted secret key generation in quasi-static environments," *IEEE Commun. Lett.*, vol. 26, no. 2, pp. 244–248, Feb. 2022.

[7] P. Staat, H. Elders-Boll, M. Heinrichs, R. Kronberger, C. Zenger, and C. Paar, "Intelligent reflecting surface-assisted wireless key generation for low-entropy environments," in *Proc. IEEE Int. Symp. Person. Indoor Mobile Radio Commun. (PIMRC)*, 2021, pp. 745–751.

[8] L. Jiao, G. Sun, J. Le, and K. Zeng, "Machine learning-assisted wireless PHY key generation with reconfigurable intelligent surfaces," in *Proc. 3rd ACM Workshop Wireless Security Mach. Learn.*, Jun. 2021, pp. 61–66.

[9] Z. Ji et al., "Secret key generation for intelligent reflecting surface assisted wireless communication networks," *IEEE Trans. Veh. Technol.*, vol. 70, no. 1, pp. 1030–1034, Jan. 2021.

[10] X. Lu, J. Lei, Y. Shi, and W. Li, "Intelligent reflecting surface assisted secret key generation," *IEEE Signal Process. Lett.*, vol. 28, pp. 1036–1040, Feb. 2021.

[11] L. Hu, G. Li, H. Luo, and A. Hu, "On the RIS manipulating attack and its countermeasures in physical-layer key generation," in *Proc. IEEE Veh. Technol. Conf. (VTC)*, 2021, pp. 1–5.

[12] Z. Wei, B. Li, and W. Guo, "Adversarial reconfigurable intelligent surface against physical layer key generation," *IEEE Trans. Inf. Forensics Security*, vol. 18, pp. 2368–2381, 2023.

[13] G. Li, A. Hu, J. Zhang, and B. Xiao, "Security analysis of a novel artificial randomness approach for fast key generation," in *Proc. IEEE Global Commun. Conf. (GLOBECOM)*, 2017, pp. 1–6.

[14] S. Jana, S. N. Premnath, M. Clark, S. K. Kasera, N. Patwari, and S. V. Krishnamurthy, "On the effectiveness of secret key extraction from wireless signal strength in real environments," in *Proc. ACM MobiCom*, Sep. 2009, pp. 321–332.

[15] X. Chen et al., "Covert communications: A comprehensive survey," *IEEE Commun. Surveys Tuts.*, vol. 25, no. 2, pp. 1173–1198, 2nd Quart., 2023.

[16] Z. Szabo, "Information theoretical estimators toolbox," *J. Mach. Learn. Res.*, vol. 15, pp. 283–287, Jan. 2014.

[17] P. Patnaik, "The non-central $\chi$2-and F-distribution and their applications," *Biometrika*, vol. 36, nos. 1–2, pp. 202–232, 1949.